GROMM goes Production

Generative Risk-neutral Option Market Models II - Production Grade

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Abstract

We refine the GROMM model presented in [WBH26] such that it has an expanding strike and expory grid, and its parameters can be estimated based on prevailing grid geometry. We compare LSTM, continuous-time GRU, and compact Mamba-style selective state-space backbones for its discrete local-volatility factor law, including variants with state-dependent spot–volatility correlation.

The resulting model has a near-perfect fit to observed option prices and outperforms comparable models in $\mathbb{P}$-measure simulations.

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1 Introduction

2 General grids

In [WBH26] GROMM was introduced for a time-homogeneous grid of strikes. Following, we introduce an interpolation matrix $\mathcal{I}^j$ that maps the density at T_{j-1} barycentrically to the strikes of maturity T_j. This interpolation matrix coincides with the one introduced in [BR15] for general grids, where the authors show that $\mathcal{I}^j$ (in their notation Ξ_j) is a transition matrix. We then demonstrate that the global existence result holds on general grids as long as each expiration has at least one interior strike and the ceiling / floor strike is an increasing / decreasing sequence across expirations.

Denote by $K^j, j \geq k$ the sequence of strikes at maturity T_j with $K^{j,0} < \cdots < K^{j,N_j} < K^{j,N_j+1} =: U^j$, N_j denoting the number of interior strikes and U^j the upper exterior strike of the j-th maturity. For such an interpolation matrix we obtain the following code book:

$$M_t^j \, q_t^j = \mathcal{I}^j \, q_t^{j-1}, \qquad M_t^j = I - \Omega_j S_t^j \ . \tag{1}$$

Grid $j-1$ has nodes $y^m := K^{j-1,m}$, grid j has nodes $K^{j,i}$. Assume every y^m lies in $[K^{j,0}, K^{j,N_j+1}]$ (automatic if both grids share the outer nodes, e.g. 0 and a cap U). If $K^{j,i} \leq y^m \leq K^{j,i+1}$, set

$$(\mathcal{I}^j)_{i,m} = \frac{K^{j,i+1} - y^m}{K^{j,i+1} - K^{j,i}}, \qquad (\mathcal{I}^j)_{i+1,m} = \frac{y^m - K^{j,i}}{K^{j,i+1} - K^{j,i}}, \tag{2}$$

and all other entries of column m zero; if $y^m = K^{j,i}$, column m is e_i. So $\mathcal{I}^j \in \mathbb{R}^{(N_j+2)\times(N_{j-1}+2)}$ has at most two non-zero entries per column.

The following lemma shows that the construction preserves the mean and unit mass:

Lemma 2.1. *(i) $\mathcal{I}^j \geq 0$, $\mathbf{1}^\top \mathcal{I}^j = \mathbf{1}^\top$ and $(K^j)^\top \mathcal{I}^j = (K^{j-1})^\top$: mass and mean are kept.*

(ii) The interpolation matrix preserves prices obtained from linear interpolation on the new strike grid. Write $C[q](K)$ for the call price of a density q at strike K. Then

$$C[\mathcal{I}^j q](K) = C[q](K) \qquad \textit{for every strike } K \textit{ of grid } j.$$

Proof. (i) The weights are non-negative by construction and by (2) $(\mathcal{I}^j)_{i,m} + (\mathcal{I}^j)_{i+1,m} = 1$. Moreover, we have for a single column m: $(\mathcal{I}^j)_{i,m} K^{j,i} + (\mathcal{I}^j)_{i+1,m} K^{j,i+1} = y^m$.

(ii)

$$\Delta_i^2 \, (y - \cdot)^+ := \frac{(y - K^{j,i+1})^+ - (y - K^{j,i})^+}{K^{j,i+1} - K^{j,i}} - \frac{(y - K^{j,i})^+ - (y - K^{j,i-1})^+}{K^{j,i} - K^{j,i-1}} = h^{j,i}(y),$$

where $h^{j,i}$ is the hat function of node $K^{j,i}$: $h^{j,i}(K^{j,i}) = 1$, $h^{j,i}(K^{j,i\pm1}) = 0$, linear in between and zero outside $[K^{j,i-1}, K^{j,i+1}]$. Hence, with $\tilde{C}(K) = \sum_m q^m \, (y^m - K)^+$,

$$\left(\Delta^2 \tilde{C}\right)_i = \sum_m q^m \, h^{j,i}(y^m) = \left(\mathcal{I}^j q\right)_i.$$

□

2.1 Existence theorem for general grids

$$dL_t^{j,i} = -\Big(\langle \xi_t^{j,i}, \eta_t^{j,i}\rangle + \tfrac{1}{2}\|\xi_t^{j,i}\|^2\Big)\,dt + \langle \xi_t^{j,i}, dW_t\rangle \tag{3}$$

In [WBH26] the existence Theorem 2.1 holds for a grid of time-homogeneous strikes across all expiries. Following we generalise this result to general lattices. For the generalised existence theorem on general grids we make the following assumption:

Assumption 2.2. (i) **Grid**: Let $\mathcal{I}^j, j > k$ define interpolation matrices such that: $K^{j,0} = 0$ for all $j \geq k$, $U^j = U^{j+1}$ for all $j \geq k$, $N_j \geq 1$ for all $j \geq k$.

(ii) **Model**:

(a) For the front, the density has mass on the interior strikes $\sum_{i=1}^{N_k} q^{k,i}_{T_{k-1}} > 0$.

(b) For all $j = k+1, \ldots, M$ and $i = 1, \ldots, N_j$, the vol-of-vol $\xi^{j,i}$ is locally Lipschitz continuous in (t, B, L) and is uniformly bounded in (B, L) by a square-integrable function in time:

$$\|\xi^{j,i}(t, B, L)\| \leq c_t \quad \text{where} \quad \int_{T_{k-1}}^{T_k} c_t^2 dt < \infty$$

where $\|\cdot\|$ is the Euclidean norm.

Assumption 2.2 (i) is made to make the theorem more concrete. In general, we only need an increasing sequence of ceilings $U^j \leq U^{j+1}$ for $j \geq k$.

Theorem 2.3 (Existence on general grids)**.** *Let $L_{T_{k-1}}$ be finite. Under Assumption 2.2 the following hold:*

(i) There is a unique strong solution for L on $[T_{k-1}, T_k)$, and every $\Sigma^{j,i}$ stays finite and strictly positive.

(ii) If spot settles at an interior strike, $J \in \{1, \ldots, N_k\}$, then L extends continuously to T_k with L_{T_k} finite.

(iii) If spot settles at an extreme strike $J \in \{0, N_k + 1\}$, then $q^j_{T_k} = e_0$ resp. $q^j_{T_k} = e_{N_j+1}$ for all $j > k$. Moreover, the remaining DLVs Σ no longer affect the model.

Proof. The proof follows [WBH26, Theorem 2.1] with the following additions: (a) a single positive interior front density node spreads, (b) η remains locally Lipschitz, (c) the calendar-like spread R is bounded on the general grid, (d) terminal settlement.

(a) By Lemma 2.1 (i) $p \coloneqq \mathcal{I}^j q^{j-1} \geq 0$ has unit mass. If q^{j-1} has interior mass, so does p: an interior node $y^m \in (0, U)$ of grid $j-1$ puts positive weight on an interior node of grid j; if $y^m < K^{j,1}$ the weight on $K^{j,1}$ is $y^m/K^{j,1} > 0$, if $y^m > K^{j,N_j}$ the weight on K^{j,N_j} is $(U - y^m)/(U - K^{j,N_j}) > 0$, and otherwise both neighbours are interior. For $j - 1 = k$ the interior mass of the front is given as in [WBH26, Theorem 2.1] under Assumption 2.2 (ii)(a), and the claim follows by induction over j. Since M^j is a non-singular M-matrix, its inverse has strictly positive interior columns and $(M^j)^{-1}e_0 = e_0$, $(M^j)^{-1}e_{N_j+1} = e_{N_j+1}$. Since $N_j \geq 1$ for all $j > k$, $q^j > 0$ if $p_i > 0$ for some $i = 1, \ldots, N_j$, and q^j has no interior mass if $p_i = 0$ for all $i = 1, \ldots, N_j$.

(b) Let $j > k$. η^j locally Lipschitz implies that L remains locally Lipschitz. The only term affected by the modified codebook (1) is $\eta_t^j = E_t^j / q_t^j$. Due to (1) we obtain for the volatility recursion

$$E^{j;f} = (M^j)^{-1}\big(\mathcal{I}^j E^{j-1;f} + \Omega_j B^{j;f} q^j\big) \ .$$

Since $\mathcal{I}^j$ is a constant matrix, $E^{j;f}$ remains a smooth function of the state, and since $q_t^j > 0$ by (a), the coefficient η_t^j remains locally Lipschitz.

(c) Moreover, we need to show that the calendar spread $R^j = \Sigma^j q^j$ defined in [WBH26, Theorem 2.2] remains a martingale. (1) gives $\Omega_j R^j = q^j - \mathcal{I}^j q^{j-1}$. Applying the discrete integration matrix Π^j with strikes defined on K^j, we obtain $\Pi^j \Omega_j R^j = C^j - C[\mathcal{I}^j q^{j-1}]$ where $\Pi^j \Omega_j$ has positive interior diagonal. Since prices are bounded on the grid, R^j remains bounded.

(d) If spot settles at an interior strike $J \in \{1, \dots, N_k\}$, the front of the next slab is $q_{T_k}^{k+1} = (M^{k+1})^{-1} \mathcal{I}^{k+1} e_J$, which has interior mass by (a), so Assumption 2.2 (ii)(a) holds on $[T_k, T_{k+1})$; L_{T_k} is finite as in [WBH26, Theorem 2.1]. If $J \in \{0, N_k + 1\}$, the shared outer nodes give $\mathcal{I}^j e_0 = e_0$ and $\mathcal{I}^j e_{N_{j-1}+1} = e_{N_j+1}$, hence $q_{T_k}^j = e_0$ resp. $q_{T_k}^j = e_{N_j+1}$ for all $j > k$, as in [WBH26, Theorem 2.1].

□

Remark 2.4 (Growing boundary strikes). In general, assumption (i) can be relaxed to a grid with growing exterior boundaries:

$$[K^{j,0}, U^j] \subseteq [K^{j+1,0}, U^{j+1}], \quad j \geq k \, .$$

In this case Theorem 2.3 (iii) is modified: a boundary strike U^k is not a boundary post settlement if $U^{k+1} > U^k$. In this case the interpolation matrix $\mathcal{I}^{k+1}$ will distribute barycentrically some of the density to the neighbouring node $K^{k+1,N_{k+1}}$. In turn, this fulfills the assumptions of

which by proof of Theorem 2.3 is going to be.

3 Factor-Law Evaluation and Conditional Arbitrage Check

3.1 Out-of-sample Factor-Law Evaluation

The historical sample is too short to justify selecting a conditional factor law from its in-sample fit. We therefore use a rolling-origin experiment. The experiment contains four consecutive test blocks of 63 business days, for a total of 252 out-of-fold observations from 13 August 2024 to 28 August 2025. Before each block, a model is fitted using only earlier observations. The number of recurrent training epochs is selected on the final 20% of that earlier sample, after which the model is refitted on the entire pre-block history.

Let $Z_t \in \mathbb{R}^F$ denote the observed factor innovation on day t, and let $C_{m,t}$ be the conditional covariance forecast by model m. The primary $\mathbb{Q}$-measure diagnostic is the Gaussian negative log score per factor and day,

$$L_{m,t}^{\mathbb{Q}} := \frac{1}{2F}\left(\log\det C_{m,t} + Z_t^\top C_{m,t}^{-1} Z_t\right). \tag{4}$$

This is a proper scoring rule for a zero-mean conditional Gaussian factor law. The original models use diagonal $C_{m,t} = \mathrm{diag}(s^2_{m,t,1}, \ldots, s^2_{m,t,F})$. To test state-dependent spot–volatility correlation without estimating an unrestricted loading matrix, we also fit

$$C_{m,t} = D_{m,t} \begin{pmatrix} 1 & \rho_{m,t}^\top \\ \rho_{m,t} & I \end{pmatrix} D_{m,t}, \qquad \rho_{m,t} = \bar{\rho} \frac{u_{m,t}}{\sqrt{1 + \|u_{m,t}\|_2^2}}, \qquad \bar{\rho} = 0.95, \tag{5}$$

where the first factor is the Gaussianized spot shock and the remaining factors are its unconditionally orthogonal DLV residuals. Hence $\|\rho_{m,t}\|_2 < \bar{\rho}$ and the covariance is positive definite by construction. Through the fixed DLV loading basis, changing $\rho_{m,t}$ changes both node-level spot–DLV correlations and the effective conditional loading directions.

Performance is reported relative to the AR(1) benchmark as the paired daily improvement

$$\Delta^{\mathbb{Q}}_m := \frac{1}{T} \sum_{t=1}^{T} \left(L^{\mathbb{Q}}_{\mathrm{AR}(1),t} - L^{\mathbb{Q}}_{m,t} \right). \tag{6}$$

Thus, positive values indicate an improvement over AR(1). Confidence intervals resample complete days using a moving-block bootstrap with ten-business-day blocks and 20,000 replications. The conclusions below are unchanged with block lengths of five or twenty business days.

For the physical drift, let $b^{\mathbb{P}}_{t,f}$ denote the estimated coefficient of the attainable $\mathbb{P}$-drift in factor direction f, let $\widehat{b}^{\mathbb{P}}_{m,t,f}$ be its forecast, and let τ_f be its training sample standard deviation. We report

$$L^{\mathbb{P}}_{m,t} := \frac{1}{F} \sum_{f=1}^{F} \left(\frac{b^{\mathbb{P}}_{t,f} - \widehat{b}^{\mathbb{P}}_{m,t,f}}{\tau_f} \right)^2. \tag{7}$$

Unlike (4), this evaluates an estimated attainable drift target. It is not a realized hedging profit-and-loss statistic.

3.1.1 LSTM, continuous-time GRU, and Mamba-style factor laws

All three recurrent models receive the same causal input x_t: the standardized current DLV-level coordinates concatenated with the previous day's factor innovation. The latter is set to zero whenever the historical panel is not contiguous. From a common hidden state, the output layer predicts the F attainable $\mathbb{P}$-drift coefficients and the F conditional log-scales. The correlated variants additionally predict the $F - 1$ unconstrained coordinates u_t in (5). Thus the models differ only in their state update, not in their information set or output parameterization.

For the LSTM, with σ denoting the logistic function, we use

$$(i_t, f_t, g_t, o_t) = \big(\sigma, \sigma, \tanh, \sigma\big) \left(W_x x_t + W_h h_{t-1} + b \right), \tag{8}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t, \tag{9}$$

$$h_t = o_t \odot \tanh(c_t), \qquad y_t = W_o h_t + b_o. \tag{10}$$

The hidden and cell states are reset at a break in the daily sequence.

The continuous-time GRU uses a conventional reset-gated candidate but replaces the usual sigmoid update gate by an exponential decay over the actual time step. Specifically,

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r), \qquad \lambda_t = \mathrm{softplus}(W_\lambda x_t + U_\lambda h_{t-1} + b_\lambda), \tag{11}$$

$$\widetilde{h}_t = \tanh(W_c x_t + U_c(r_t \odot h_{t-1}) + b_c), \qquad a_t = \exp(-\lambda_t \Delta t), \tag{12}$$

$$h_t = a_t \odot h_{t-1} + (1 - a_t) \odot \widetilde{h}_t, \qquad y_t = W_o h_t + b_o. \tag{13}$$

Here $\Delta t = 1/252$ on the daily panel. The annual decay is initialized at 25, corresponding to one-day retention $\exp(-25/252) = 0.906$, and is then learned as a positive state-dependent quantity.

The model denoted "Mamba" in the experiments is a compact Mamba-style selective state-space recurrence rather than a full stacked Mamba3 network. Its input-dependent retention gate and candidate state are

$$a_t = \sigma(W_a x_t + b_a), \qquad \widetilde{h}_t = \tanh(W_c x_t + b_c), \tag{14}$$

$$h_t = a_t \odot h_{t-1} + (1 - a_t) \odot \widetilde{h}_t, \qquad y_t = W_o h_t + b_o. \tag{15}$$

This has fewer recurrent interactions than the LSTM and provides a useful small-sample test of whether input-dependent memory is sufficient. All three models use 32 hidden coordinates. Their output vector is partitioned as

$$y_t = \left(\widehat{b}_t^{\mathbb{P}}, \ell_t, u_t\right), \qquad s_{t,f} = \bar{s}_f \exp(\mathrm{clip}(\ell_{t,f}, -\log 3, \log 3)), \tag{16}$$

where the final block is omitted in the diagonal models. On each fitting prefix we minimize

$$\frac{1}{T_{\text{train}}} \sum_t L_{m,t}^{\mathbb{Q}} + \frac{1}{T_{\text{train}}} \sum_t L_{m,t}^{\mathbb{P}} + \lambda_\rho \frac{1}{T_{\text{train}}} \sum_t \|\rho_{m,t}\|_2^2, \qquad \lambda_\rho = 10^{-2}, \tag{17}$$

using Adam. The correlation penalty is zero for the diagonal variants. Training duration is selected using only the pre-test validation suffix and the selected number of epochs is then used to refit the complete pre-test history.

Table 1: Rolling-origin factor-law evaluation. Lower mean scores are better. Improvements and confidence intervals are measured relative to AR(1); positive improvements are favourable. The reported $\|\rho_t\|_2$ is a latent loading norm, not an observable spot–volatility correlation.

	$\mathbb{Q}$ factor covariance				Attainable $\mathbb{P}$ drift		
Model	Mean score	Improvement	95% interval	Mean $\|\rho_t\|_2$	Mean error	Improvement	95% interval
Fixed	0.5258	−0.0106	[−0.0258, 0.0003]	0	0.7342	−0.3270	[−0.4153, −0.2478]
AR(1)	0.5152	—	—	0	0.4072	—	—
LSTM	0.5063	0.0088	[−0.0471, 0.0658]	0	**0.2722**	**0.1350**	[**0.0993**, **0.1682**]
LSTM with ρ_t	0.6125	−0.0973	[−0.3397, 0.0475]	0.9007	0.2927	0.1145	[0.0663, 0.1572]
Continuous-time GRU	0.4826	0.0325	[−0.0118, 0.0938]	0	0.3388	0.0684	[0.0135, 0.1148]
Continuous-time GRU with ρ_t	0.4994	0.0157	[−0.0713, 0.0844]	0.8852	0.3629	0.0443	[−0.0060, 0.0852]
Mamba SSM	0.4859	0.0292	[−0.0229, 0.0794]	0	0.3484	0.0588	[0.0100, 0.1023]
Mamba SSM with ρ_t	**0.4796**	**0.0355**	[−0.0476, 0.1096]	0.8682	0.3401	0.0671	[0.0254, 0.1061]

The correlated Mamba-style selective state-space model has the best average $\mathbb{Q}$ score. The diagonal continuous-time GRU is close: its score of 0.4826 is slightly better than the diagonal Mamba score of 0.4859, but their paired improvement is only 0.0033, with a 95% interval of $[-0.0483, 0.0663]$. The GRU therefore provides a competitive continuous-time memory parameterization, not a statistically established improvement. Relative to its diagonal version, however, its mean improvement is only 0.0063, with a direct paired 95% interval of $[-0.0429, 0.0484]$. The extra correlation state is therefore learnable but does not yet deliver a statistically established predictive gain. The correlated LSTM is materially worse in $\mathbb{Q}$, especially in the third test block, which is direct evidence of overfitting. The LSTM improvement in the attainable $\mathbb{P}$-drift target remains the clearest result. The GRU correlation head also deteriorates its diagonal model: the direct $\mathbb{Q}$ improvement is -0.0168, with a 95% interval of $[-0.1325, 0.0637]$, and its attainable-$\mathbb{P}$

error increases from 0.3388 to 0.3629. Thus an exponential memory gate alone does not resolve the spot-correlation overfitting problem. These diagnostics evaluate the dynamics model; final model selection must be based on out-of-sample hedging performance for economically relevant portfolios.

3.1.2 Observable spot–ATM-volatility correlation

The factor score evaluates the latent diffusion, whereas a directly observable leverage diagnostic is the expiry-dependent correlation

$$\rho_{S,\mathrm{ATM}}(\tau) := \mathrm{Corr}\left(\frac{\Delta S_t}{S_{t-1}}, \Delta \log \sigma_{\mathrm{ATM},t}(\tau)\right), \tag{18}$$

where spot returns are forward-adjusted and $\sigma_{\mathrm{ATM},t}(\tau)$ is the Black implied volatility at a fixed rolling business-day tenor τ. The historical curves in Figure 1 use the published SANOS ATM term structure. The full curve contains 890 daily transitions from January 2022 to August 2025; the recent curve contains the 258 transitions from 13 August 2024 onward.

For the model curves, we start from the calibrated state on 29 August 2025 and simulate one business day under $\mathbb{Q}$ using eight DLV substeps and 5,000 paths arranged as 2,500 antithetic pairs. The constant model keeps the unconditional factor loadings, while the Mamba-style selective SSM applies its state-dependent factor amplitudes. At the next date, ATM prices are inverted to Black volatilities at the same rolling tenor. When that tenor lies between two surviving fixed expiries, total variance is interpolated linearly. The 125D simulation point is omitted because a one-day rolling 125D volatility requires the birth of an expiry beyond the terminal model date.

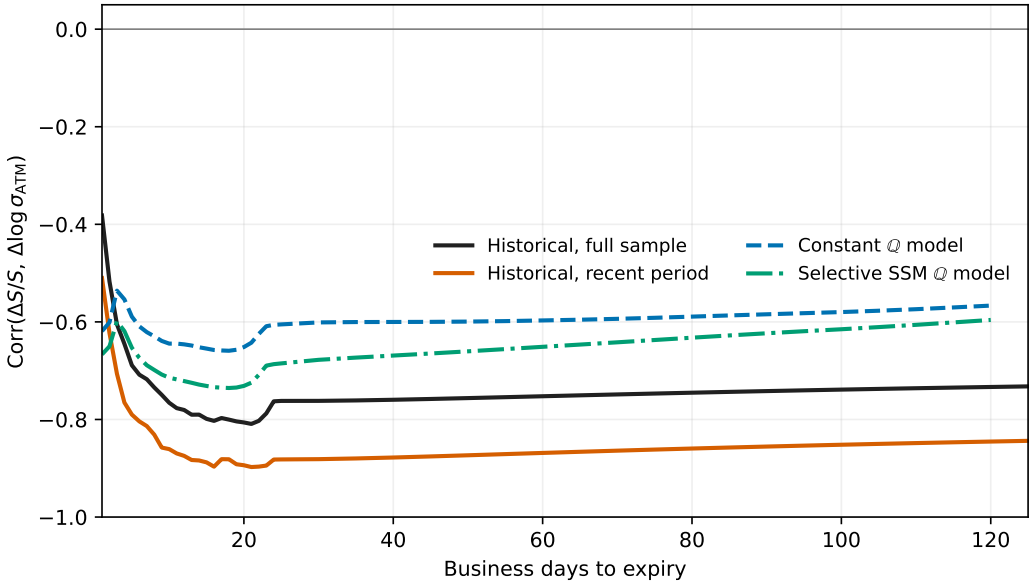


Figure 1: Term structure of daily forward-adjusted spot-return correlation with changes in rolling-tenor ATM Black implied volatility. Historical curves are measured under $\mathbb{P}$; model curves are one-day conditional simulations under $\mathbb{Q}$ from the final calibration state.

Table 2: Selected points from Figure 1. The model columns are total observable correlations, including the DLV and 5% Black background contributions.

Business days	Historical	Recent	Constant $\mathbb{Q}$	Mamba SSM $\mathbb{Q}$
1	−0.382	−0.510	−0.619	−0.667
5	−0.689	−0.790	−0.589	−0.653
10	−0.766	−0.861	−0.645	−0.714
20	−0.806	−0.894	−0.652	−0.731
30	−0.762	−0.881	−0.601	−0.678
60	−0.752	−0.869	−0.597	−0.651
90	−0.742	−0.856	−0.584	−0.623
120	−0.733	−0.845	−0.567	−0.596

The selective backbone improves the root-mean-square discrepancy from the full historical curve from 0.148 to 0.098. It is close to the full-sample 5D correlation (−0.653 versus −0.689), but does not reproduce the stronger recent value of −0.790. More generally, it is too negative at 1–3D and too weak beyond roughly one month. The corresponding discrepancy from the recent curve is 0.188, compared with 0.246 for the constant model. Thus the selective state improves the observable leverage effect, but the maturity shape and recent regime remain material targets for the state-dependent correlation head in Equation (5).

The 5D gap has a more specific mechanical explanation. During estimation, the first factor is the Gaussian score of the observed product-spot distribution, including the 5% Black background. In the current simulator, that factor settles the residual front density, after which the Black background is sampled independently. This preserves the two marginal laws but not the fitted product-spot copula. On 29 August 2025 the one-day ATM volatility is 9.56%, so the independent 5% background contributes 27.4% of one-day spot variance. In the corresponding first-order decomposition it attenuates spot correlation by

$$\sqrt{1 - \frac{0.05^2}{0.0956^2}} = 0.852. \tag{19}$$

Applying this attenuation to the recent 5D historical correlation gives $-0.790 \times 0.852 = -0.673$, close to the simulated value of −0.653. The principal 5D discrepancy is therefore the independent product-lift coupling, not the choice between Mamba and GRU memory. A corrected simulator should either bridge the observed product spot directly and sample the latent components conditionally, or estimate and simulate a genuinely latent spot score. The learned ρ_t head must then be propagated into the path covariance; it is not used by the present constant/selective curves. The remaining short- and long-maturity shape differences are separate loading-basis and state-dependence questions.

3.1.3 Background-factor identification and estimates

The product representation does not identify a unique latent background process. At one expiry, many joint laws of (X_T, Y_T) produce the same law of $S_T = X_T Y_T$. Option prices therefore identify the product law but not the allocation of its risk between the discrete factor and the smoothing factor. We distinguish three specifications rather than treating them as mutually identified estimates.

First, the production baseline fixes

$$\frac{\mathrm{d}Y_t}{Y_t} = \sigma_Y \, \mathrm{d}B_t, \qquad \sigma_Y = 5\%, \tag{20}$$

with B independent of the DLV factors. This is a modeling convention, not an estimated process. It retains the simple Black convolution and the current closed risk-neutral drift equations.

Second, the released SANOS state provides an observable model-component proxy for a floating background: its Black-kernel total-variance curve. We retain the published smoothness multiplier and interpolate total variance to the GROMM2 expiry grid. Figure 2 summarizes its history. Median annualized kernel volatility is between 7.35% and 7.57% across the grid, compared with the fixed 5% convention, and its 10–90% range at 5D is 4.86%–13.20%. The first kernel-volatility factor explains 77.6% of daily log-change variance; two factors explain 90.3%, four explain 96.3%, and eight explain 98.6%.

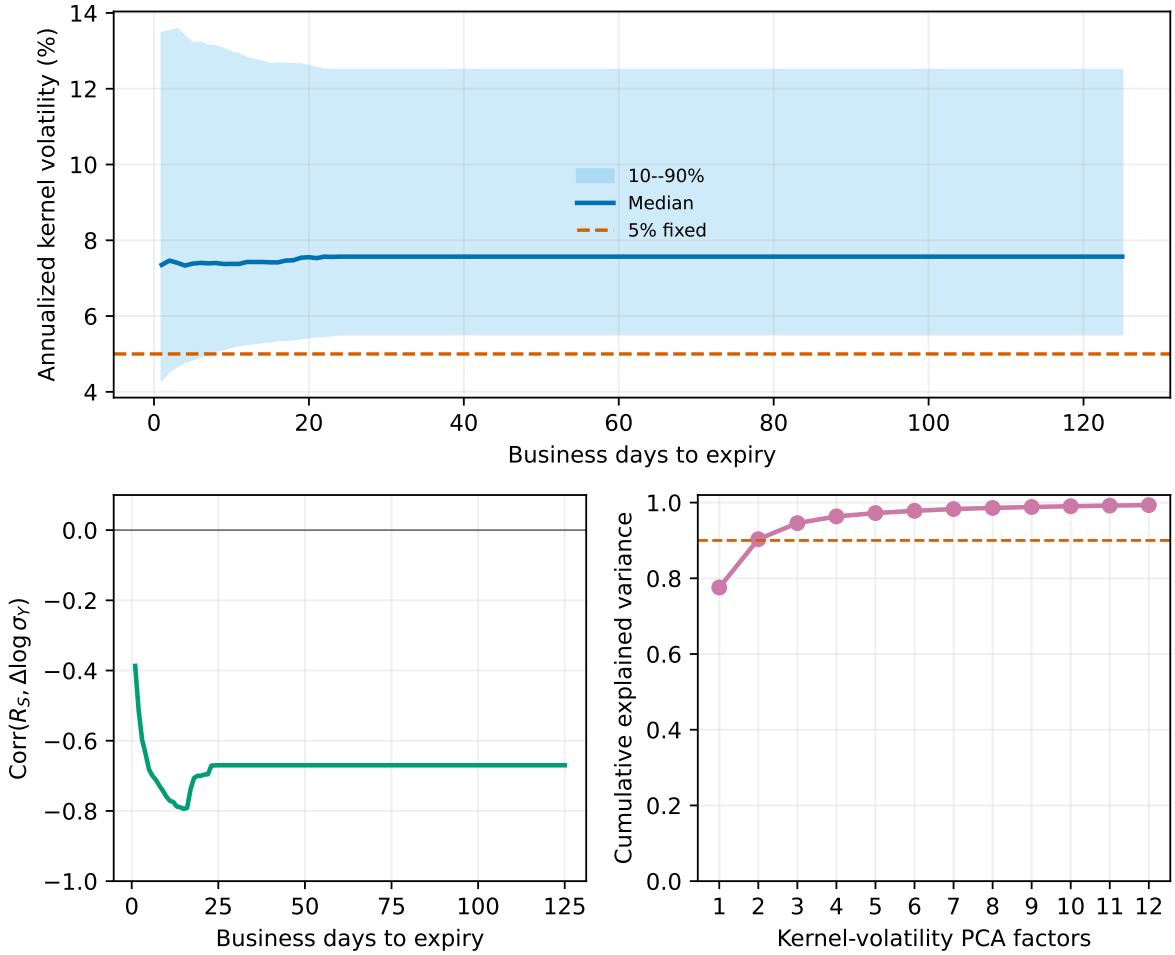


Figure 2: Historical estimates for the released SANOS Black-kernel process. Left: the kernel-volatility term structure. Centre: its daily correlation with forward-adjusted spot returns. Right: cumulative PCA power of daily log-kernel-volatility changes. Beyond the released terminal kernel node, the curve follows the documented flat annualized-volatility extrapolation.

Third, as a parsimonious stochastic-volatility benchmark, we fit under $\mathbb{P}$ the one-factor CIR/Heston variance law [Hes93]

$$\mathrm{d}v_t = \kappa_P(\theta_P - v_t)\,\mathrm{d}t + \nu\sqrt{v_t}\,\mathrm{d}B_t^v \tag{21}$$

to the released one-business-day kernel variance, using the exact CIR conditional first two moments in a Gaussian quasi-likelihood. The estimates are $\kappa_P = 19.74$, $\sqrt{\theta_P} = 9.36\%$, and $\nu = 0.935$. The Feller ratio $2\kappa_P\theta_P/\nu^2$ is 0.395, so a positivity-preserving scheme would be required. The standardized variance innovation has correlation -0.320 with the observed product-spot score; a ten-day moving-block bootstrap gives the 95% interval $[-0.413, -0.207]$. This last number is a reduced-form observable correlation, not an identified correlation between the unobserved Y return and its variance shock.

We also test an independent Heston background directly in the surface calibration. On 29 August 2025, a Heston model is fitted to the SANOS surface, scaled to 5% instantaneous volatility, and its correlation is set to $\rho = -0.612$. The resulting background has material downside skew. At representative strikes $K_z = \exp\{z\,9.56\%\sqrt{T}\}$, its left-minus-right implied-volatility spread between $z = -1$ and $z = 1$ is 2.12, 2.59, 2.61, and 2.51 volatility points at 5D, 20D, 60D, and 120D, respectively. This is a reallocation of the observed skew between the background and residual DLV components: recalibration still targets the same total option surface.

Table 3 records the fit evidence. The one-date control already shows the central issue: under the same residual-DLV extraction, the scaled Heston background loses 6.13 percentage points of inside-spread fit relative to the 5% Black background. The Heston background is not close to Black after scaling: preserving its relative variance-of-variance together with $\rho = -0.612$ imposes substantial downside skew. The residual DLV would have to deconvolve this shape, but a non-negative local variance cannot implement an arbitrary signed deconvolution. On the reference date, 3.86% of residual DLV nodes reach the lower variance floor under Heston, versus 2.13% under Black; neither specification reaches the upper cap.

The rolling experiment uses the reference-date Heston parameters at every historical date and changes only the initial variance. It therefore also imposes one skew and term-structure family on all historical surfaces. The date-varying FFT-kernel interpolation has maximum validated absolute price error 2.7×10^{-5}, so interpolation alone cannot explain the gap. The historical Heston run in the table predates the current symmetrized M-matrix solver and is consequently not a controlled comparison with the current-solver Black result. A reduced-learning-rate continuation made only 69 current-solver steps before the 50-epoch plateau-review rule stopped it; it did not improve materially. A full current-solver Heston refit remains necessary before drawing a long-history quantitative conclusion.

Table 3: Evidence on the independent Heston background. Static controls use the 29 August 2025 SANOS surface; historical results use the six-month, 45-expiry synthetic SANOS book. Errors are measured in synthetic half-spreads.

Experiment	Background	Inside spread	Mean excess
One-date control	Black, fixed 5%	95.10%	0.058
One-date control	Heston, scaled 5%	88.97%	0.228
Historical, current solver	Black, fixed 5%	91.22%	0.078
Historical, old solver	Heston, date-varying v_0	83.11%	0.270
Historical, current solver	Heston continuation (69 steps)	83.30%	0.269

More importantly, the rolling extraction does not separate Heston variance from residual DLV risk. After removing the estimated front factor, the Heston variance score still explains 23.0% of residual DLV innovation energy. Its largest absolute correlations with the retained DLV factors are 0.767 in training and 0.668 on the chronological holdout. Thus the Heston kernel adds static skew, but this particular decomposition supplies no evidence of a cleaner dynamic spot–volatility representation.

Table 4: Status of the background specifications. "Estimated" refers to historical $\mathbb{P}$ estimation; none of the stochastic alternatives yet has a separately identified $\mathbb{Q}$ drift.

Specification	Historical result	Risk-neutral status
Independent Black	Fixed $\sigma_Y = 5\%$; not estimated	Current closed construction
Floating SANOS kernel	Median 7.35–7.57%; two PCA factors explain 90.3%	Variance-risk premium not identified
CIR/Heston proxy	$(\kappa_P, \sqrt{\theta_P}, \nu) = (19.74, 9.36\%, 0.935)$	$\mathbb{Q}$ variance drift and latent leverage not identified
Independent Heston pilot	Adds 2.1–2.6 volatility points of downside background skew; one-date fit falls from 95.10% to 88.97%	Residual DLV must counteract the imposed skew; full current-solver historical refit outstanding

The floating-kernel and Heston estimates are therefore informative $\mathbb{P}$-models of a background state, but neither experiment identifies the background market price of variance risk separately from the DLV surface. The risk-neutral constructions considered below retain the fixed independent Black background.

3.2 Conditional Arbitrage Check

The check starts from an empty portfolio, with no target liability. Let R_{t+1} be the vector of next-day discounted instrument gains and let $a_t = \pi_\theta(X_t, h_t)$ be predictable holdings, where X_t contains only information available before trading and h_t records the current holdings. The strategy's terminal trading gain is

$$G_\theta := \sum_{t=0}^{T-1} a_t^\top R_{t+1}. \tag{22}$$

For risk aversion $\lambda > 0$, the frictionless entropic diagnostic minimizes

$$L_{\mathrm{ent}}(\theta) := \frac{1}{\lambda} \log \mathbb{E}^{\mathbb{Q}}[\exp(-\lambda G_\theta)]\,. \tag{23}$$

Under the conditional martingale null $\mathbb{E}^{\mathbb{Q}}[R_{t+1} \mid \mathcal{F}_t] = 0$, every integrable predictable strategy has zero expected gain. Jensen's inequality then gives $L_{\mathrm{ent}}(\theta) \geq 0$, and the zero-action strategy attains zero. An improvement significantly below zero on independent evaluation paths is evidence against this null within the chosen strategy class, not a proof of a sure-profit arbitrage. Nonnegative trading costs may be subtracted from G_θ, but make the diagnostic less sensitive to small drift defects.

Training and evaluation paths must remain separate. After fitting, freeze the policy and evaluate its mean trading gain and risk-adjusted performance against the zero-action baseline. For uncertainty estimates, retain whole paths as sampling units and average antithetic partners before

computing standard errors. If validation paths are used for checkpoint selection or early stopping, report validation-selected performance rather than an untouched significance test. Hedging or pricing a nonzero target payoff is a separate experiment and is not part of this conditional arbitrage check.

GPU linear algebra. We solve the same float64 tridiagonal systems by parallel cyclic reduction on the GPU, retaining the sequential solver as a reference. On an RTX 5090, a warmed batch of 256 complete twenty-day paths, with eight Brownian substeps per day, takes 3.23 seconds instead of approximately thirteen minutes with sequential elimination. A complete-path comparison using identical Gaussian draws gives option-price differences below 1.5×10^{-15}, identical forward-start straddle payoffs, and mass and forward-mean errors below 7×10^{-14}. The model parameters, grid and time steps are unchanged. These checks establish numerical equivalence of the implementation; they do not replace the unconditional and held-out strategy-conditional martingale diagnostics.

4 Deep Hedging of a Forward-Start Straddle

4.1 Experiment and Hedge Books

We compare two completed daily hedging experiments for the cash payoff

$$H = |S_{20}/S_{10} - 1|, \tag{24}$$

with initial normalized spot $S_0 = 1$. Both use the same one million training paths and one million independent validation paths from the saved GROMM2 $(-3, 2)$ affine-200, 16-factor AR(1) model, initialized on 29 August 2025. This is not the newer $(-4, 3)$ rolling calibration. The Black background volatility is set to zero in the simulator, without refitting the historically estimated DLVs or factor law, which used a 5% background. There are eight Brownian substeps per business day, but trading occurs only daily. Paths retain their original equal $\mathbb{Q}$ weights: no recentering or empirical change of measure is applied.

The two hedge books are:

1. **Ordinary options and spot (85 controls).** Fourteen contracts have fixed inception-relative strikes $\{0.95, 0.98, 0.99, 1, 1.01, 1.02, 1.05\}$ and expiries on days 10 and 20; they can be rebalanced daily before expiry. On each day 1 through 10, seven additional options are issued at those strikes relative to that day's spot, with ten business days to expiry. These cohorts are held to cash settlement, except that the day-10 ATM call expiring on day 20 remains tradable on days 10 through 19. Puts are used below ATM and calls at or above ATM, with the side fixed at issuance. The day-zero cohort duplicates the fixed ten-day contracts and is not counted twice. Spot can be traded daily.

2. **PCA baskets, spot and a one-day ATM call (17 controls).** Fifteen PCA-mimicking portfolios are formed at inception from 44 vanilla contracts with expiries of at least 25 business days. Their constituent weights remain fixed, while basket holdings can change daily. Spot and a newly issued one-day ATM call are also tradable each day. This book does not contain the day-10 ATM call expiring on day 20.

Transaction costs are 0.1 basis points of absolute unit turnover, without an S_t multiplier. PCA costs are charged per basket, not per underlying leg. Option cash settlement is free; spot and basket liquidation is charged.

4.2 Objective and Price Convention

For risk aversion $\lambda = 100$, we use the power-exponential utility and its optimized certainty equivalent

$$u_\lambda(x) = \begin{cases} (1 - e^{-\lambda x})/\lambda, & x \geq 0, \\ x - \lambda x^2/2, & x < 0, \end{cases} \qquad \mathrm{OCE}_\lambda(X) := \sup_{m \in \mathbb{R}} \{m + \mathbb{E}^{\mathbb{Q}}[u_\lambda(X - m)]\}. \tag{25}$$

Let G_θ denote terminal trading gains after costs. Separate policies are trained for the empty, short-claim and long-claim portfolios, with no shared trainable parameters. The risks and indifference quotes are

$$\rho_0 = \inf_\theta[-\mathrm{OCE}_\lambda(G_\theta)], \qquad \rho_- = \inf_\theta[-\mathrm{OCE}_\lambda(G_\theta - H)],$$
$$\rho_+ = \inf_\theta[-\mathrm{OCE}_\lambda(G_\theta + H)], \qquad p_{\mathrm{ask}} = \rho_- - \rho_0, \quad p_{\mathrm{bid}} = \rho_0 - \rho_+. \tag{26}$$

The reported values approximate these optimizations within the chosen policy class. Each policy combines an affine and a two-layer deep action head with a 32-dimensional LSTM state. Its information includes time, latent model state, current option prices and inventory. Positions have a numerical cap of ten units and a soft near-cap penalty. Training uses batches of 256 paths; validation selects checkpoints and stops each policy after 50 epochs without sufficient improvement, with learning-rate reductions after 25 stagnant epochs. The no-trading policy is retained as an epoch-zero candidate.

4.3 Results

Table 5 reports the final validation-selected policies on both complete samples.[1] Antithetic partners remain together in each sample, giving 500,000 independent pairs per set. The displayed errors are three plug-in Monte Carlo standard errors for frozen policies, including covariance when forming price differences. They exclude fitting and checkpoint-selection uncertainty and are marginal error bounds, not a simultaneous coverage guarantee. Validation is used for model selection, not as a third, untouched test sample.

Table 5: Forward-start straddle prices, in percent of initial spot. Every entry is an estimate ± three antithetic-pair Monte Carlo standard errors. Transaction costs are included.

Quantity	Training	Validation
$\mathbb{E}^{\mathbb{Q}}[H]$	1.7621 ± 0.0085	1.7566 ± 0.0082
Ordinary-book bid	1.0623 ± 0.0025	1.0627 ± 0.0025
Ordinary-book ask	2.4457 ± 0.0246	2.8092 ± 0.6186
PCA-book bid	1.5308 ± 0.0017	1.5006 ± 0.0053
PCA-book ask	2.0300 ± 0.0028	2.0724 ± 0.0193
Empty-portfolio risk, both books	0	0

The ordinary-book run stopped after 54 epochs, selecting epochs 0, 4 and 0 for the empty, short and long portfolios respectively. The PCA run stopped after 63 epochs, selecting epochs 0,

[1] Run identifier: `spot-pca-sequential-20261029-v1`; the archived `quotes.json` and `training_report.json` files record each book's prices, sampling errors, model provenance and stopping decisions.

8 and 13. Thus both empty portfolios select no trading; the ordinary-book buyer also selects the unhedged baseline. The PCA book narrows the validation bid–ask spread from 1.7466 to 0.5718 percentage points, with the Monte Carlo expectation inside both brackets. Its three-standard-error expanded validation bracket is $[1.4953\%, 2.0917\%]$, compared with $[1.0601\%, 3.4278\%]$ for the ordinary book. The ordinary seller's much larger error indicates sensitivity to rare losses.

These are learned, daily hedging results for different trading universes, not exact replication bounds or a proof of risk neutrality. No-trading selection only says that training did not find a validation-improving empty-portfolio strategy. Finite-step martingale defects remain a separate model diagnostic. Instantaneous factor spanning does not imply perfect replication with daily trading, fixed basket weights, costs and position limits.

5 VIX Reconstruction

We compare the published Cboe VIX close with an end-of-day market reconstruction and the corresponding GROMM2 prices. This experiment uses the earlier six-month affine-200 history, with 200 interior strikes over NMS $(-3, 2)$ and an independent 5% Black background; it is not a result for the ongoing rolling Gromm v2 4,3 calibration. The sample spans 3 January 2022 to 29 August 2025. All statistics below use the same 882 matched dates out of 891 observations; nine dates lack eligible expiries bracketing 30 calendar days.

For each date, we select the two eligible Friday expiries around 30 calendar days and apply the discrete-strike VIX calculation.[2] The observed option book determines the implied forward, the strike K_0 immediately below it, and the out-of-the-money put and call strips. Zero-bid options are excluded, and each wing stops after two consecutive zero bids. The price at K_0 averages its call and put. GROMM2 reprices precisely these contracts, retaining the same forwards, discount factors and maturity weights as the market reconstruction; we do not integrate over all model strikes. AM and PM settlement clocks are retained. The reconstruction uses OptionMetrics end-of-day quotes and its cached zero curve rather than Cboe's contemporaneous constituent snapshot and official rate inputs.

Writing D_j for the discount factor, F_j for the forward and $O_j(K_i)$ for the selected option price at expiry T_j, its total variance is

$$w_j = \frac{2}{D_j} \sum_i \frac{\Delta K_i}{K_i^2} O_j(K_i) - \left(\frac{F_j}{K_{0,j}} - 1 \right)^2 . \tag{27}$$

For calendar-year maturities $T_1 \le \tau < T_2$, where $\tau = 30/365$,

$$\mathrm{VIX} = 100 \sqrt{\frac{1}{\tau} \frac{(T_2 - \tau) w_1 + (\tau - T_1) w_2}{T_2 - T_1}} . \tag{28}$$

We also show an analytic terminal-marginal variance proxy, computed without Monte Carlo. At model time $\tau_* = 30/365$, let p_i be the latent probabilities on support k_i and let $\sigma_B = 0.05$. Independence of the Black background gives

$$V_{\mathrm{marg}} := 100 \sqrt{\frac{e^{\sigma_B^2 \tau_*} \sum_i p_i k_i^2 - \left(\sum_i p_i k_i \right)^2}{\tau_*}} . \tag{29}$$

[2] See the Cboe Volatility Index Methodology, `https://cdn.cboe.com/resources/vix/VIX_Methodology.pdf`, accessed 26 October 2026.

This is the annualized standard deviation of the terminal normalized simple return, not the VIX log-contract statistic. It uses a single model horizon rather than the two actual settlement clocks.

Table 6: VIX comparison on 882 matched dates. Bias is model minus Cboe VIX close; error columns are annualized volatility points.

Series	Bias	Median absolute error	RMSE	Correlation
Market reconstruction	+0.026	0.087	0.255	0.9991
GROMM2, same strips	+3.170	2.592	3.875	0.9965
GROMM2 marginal proxy	+1.205	0.711	2.251	0.9898

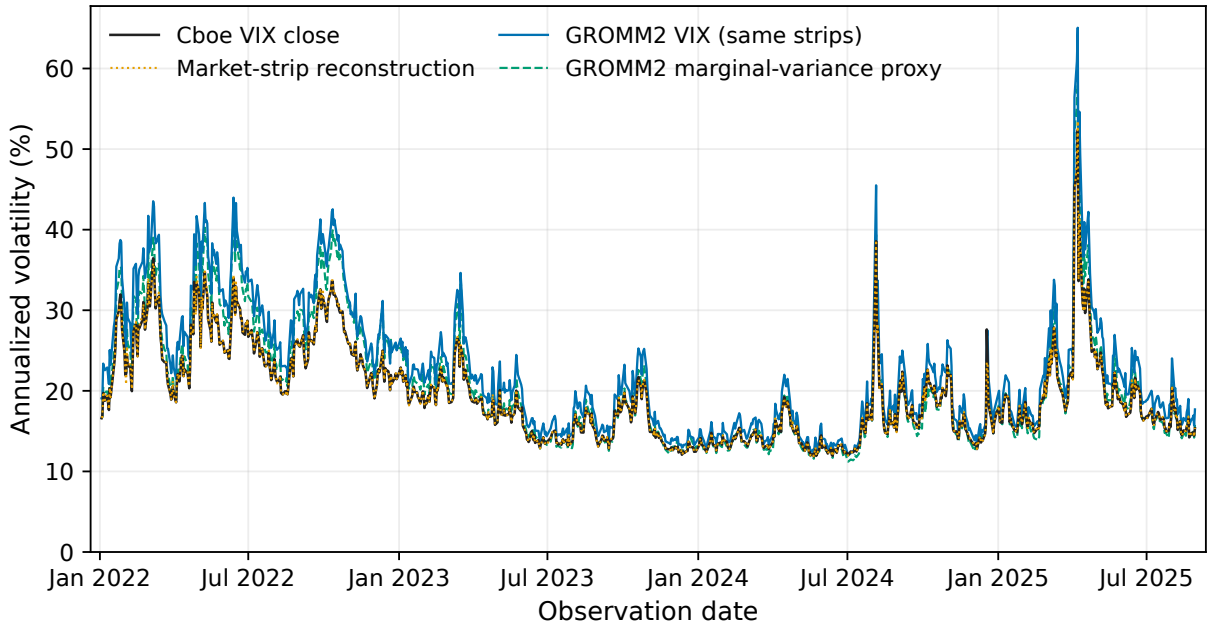


Figure 3: Published VIX, the market-strip reconstruction, GROMM2 repricing of the same strips, and the analytic terminal-marginal variance proxy. All curves use the matched dates in Table 6.

The market reconstruction closely tracks the published close. The earlier GROMM2 calibration substantially overstates the VIX level despite strong co-movement. The marginal proxy is numerically closer, but it is a different statistic and cannot replace the VIX comparison. This is a surface-fit diagnostic, not a dynamic no-arbitrage test.

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